

Junhao Zhang

Urbana, IL • (507) 319-9832 • jzhang1707@outlook.com • [/in/jzhang2029/](https://in/jzhang2029/) • jzhang.dev

EDUCATION

University of Illinois Urbana-Champaign

Bachelor of Science in Mathematics & Computer Science

Bachelor of Art in Spanish

Expected Graduation: May 2029

Cumulative GPA: 3.98/4.00

SKILLS

Technical Skills: C/C++, Rust, TypeScript, React, Node.js, Java, Python, Git, LaTeX

Languages: English (Native), Chinese (Native), Spanish (Advanced Conversational Proficiency)

RELEVANT COURSEWORK

CS 225 (Data Structures & Algorithms in C++), **CS 128** (Intro to C++), **CS 128H** (Intro to Rust), **CS 357** (Numerical Methods)

Math 314 (Introduction to Higher Mathematics), **Math 257** (Linear Algebra with Computational Applications)

EMPLOYMENT AND RESEARCH

Cloudflare | Software Engineer Intern | Austin, TX

May 2026 – Present

- Migrate and extend an internal WARP client debugging tool in React and Rust/WASM to reduce investigation time
- Integrate Cloudflare Workers AI to enable AI-assisted issue diagnosis while preserving customer privacy
- Author a functional specification and collaborate with a PM to scope and align project requirements

CS128H | Course Assistant | Urbana, IL

January 2026 – Present

- Host weekly office hours and provide one-on-one tutoring to clarify course concepts
- Design lecture presentations, and homework assignments for 60+ students in an introductory Rust course
- Collaborate with instructors to refine course content and improve learning outcomes

Paleopollen Research Lab | Student Researcher | Urbana, IL

January 2026 – Present

- Develop two-stage SAM2 (Segment Anything Model)-based pipeline to automatically segment pollen grains
- Design heuristic-driven prompting strategies resulting in accurate segmentation of all test data (30 samples)
- Process large microscopy datasets using CZI extractor and ndpi-tile-cropper on NCSA cloud compute
- Present results in research poster at the Undergraduate Research Symposium with 750+ participants

PROJECTS

RusTX | github.com/pranavpopuri/raytracing-in-rust | Urbana, IL

November 2025 – December 2025

- Led a team of 3 to design and implement a ray tracer in Rust as a CS128 Honors final project
- Applied Rust's concurrency features to create a multi-threaded renderer reducing render time from 10 minutes to 2 minutes

Europa Lang | github.com/cursorweb/europa-lang | Rochester, MN

September 2021 - Present

- Implemented a dynamic programming language in Rust with a recursive descent parser
- Led a community of 4 maintainers, overseeing code reviews and feature proposals

FotoFindr (HackIllinois) | github.com/slothinthehat/fotofindr | Urbana, IL

March 2026

- Developed a mobile app in team of 3 that indexes camera roll for natural-language search, and flags low-value images for deletion
- Built a full-stack system with Expo React Native (TypeScript), FastAPI, YOLO, DeepFace, Gemini API, and Snowflake

ACTIVITIES

University of Illinois Symphony Orchestra | Violinist | Urbana, IL

August 2025 – Present

- Perform in an Orchestra of music performance majors as well as highly talented non-majors
- Hold concerts 3 times per semester, with a variety of repertoire including works by Brahms, Shostakovich, Mahler, and Beethoven

Volunteer Income Tax Assistant (VITA) | Volunteer | Urbana, IL

January 2026 – Present

- Contribute to the completion of 300+ tax returns in the community by helping verify eligibility for VITA services
- Provide Spanish-English translation, supporting community members via phone

HONORS AND AWARDS

Dean's List

January 2026

Minnesota Bilingual Seal (Spanish and Chinese) - Platinum Level

2024, 2025

- Demonstrate ACFTL advanced low proficiency in Speaking, Writing, Reading, and Listening in Chinese and Spanish